14/2/24 (AN)

Hall Ticket Number:

MR-21

Code: B414CB5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, Jan/Feb-2024

Software Engineering

(Computer Science and Business System)

Time: 3 Hours

Max. Marks: 70

Answer ALL Questions

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define Software myth. List out the customer myths.	2	1	1	2
1. b)	Enumerate different layers in Software Engineering.	2	1	1	2
1. c)	Differentiate user and system requirements.	2	3	2	2
1. d)	What is a feasibility study?	2	1	2	1
1. e)	Determine how design produces quality for a given software.	2	3	3	2
1. f)	What is modularity? How it affects a given software?	2	2	2	1
1. g)	What is a unit testing?	2	1	4	2
1. h)	What is a metric for a given software?	2	1	4	1
1. i)	What is software reliability?	2	1	4	1
1. j)	List out different types of risks.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Demonstrate why software engineering is a layered technology.	3	3	1	2
2. b)	Elaborate different incremental process models.	7	2	1	1
	OR				
2. c)	Explain Capability Maturity Model Integration	3	1	1	1
2. d)	Draw and Explain prototyping model with a neat sketch.	7	3	1	2
3. a)	Differentiate functional and non-functional requirements.	4	4	2	2
3. b)	Illustrate how context modelling helps to model a system.	6	1	2	1
	OR				
3. c)	Examine the structure of Software Requirements document.	4	4	2	2
3. d)	What is object modelling of a system? Explain with an example.	6	6	2	1
4. a)	List out any 4 design concepts and briefly explain them.	4	4	3	2
4. b)	Explain about different types of relationships a UML model can contain?	6	2	3	1
	OR				
4. c)	What are different levels of design a design model presents? Explain.	4	1	3	1
4. d)	Develop use case diagram for online course registration system.	6	6	3	2
5. a)	Distinguish different methods involved in white box testing.	6	4	4	2
5. b)	Explain the metrics for analysis and design model.	4	1	4	1
	OR				
5. c)	Explain validation and system testing in detail.	5	1	4	1
5. d)	Analyze the different source code and testing metrics.	5	4	4	1
6. a)	Differentiate reactive vs proactive risk management.	4	1	5	1
6. b)	What are different activities performed in software quality assurance?	6	2	5	2
	OR				
6. c)	Explain how to identify and project a risk.	4	1	5	1
6. d)	What are software reviews? Explain how a formal technical review is conducted.	6	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome